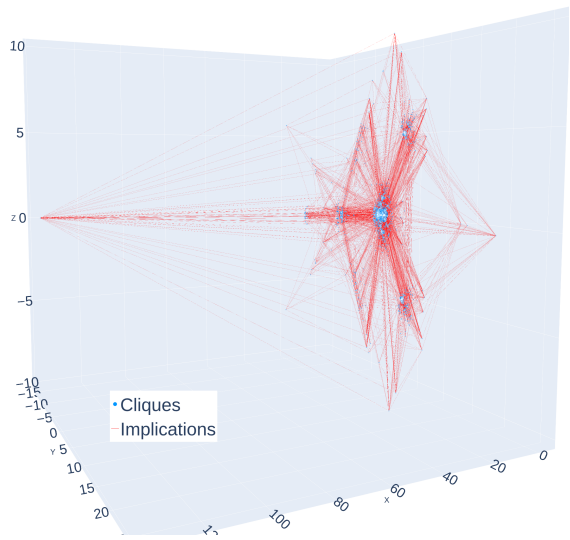

Figure 7: A global view of the latent space of equational theories and of the 1415 vertices and 4824 edges of the implication graph $(G, \rightsquigarrow)$ modulo reversibility. The existence of implication flows reveals an emerging similitude between the model-theoretic definition of the latent space and the proof-theoretic definition of the implication graph.

consequence: it enables us to derive from the original embedding of $(G, \rightsquigarrow)$ an embedding of the graph $(G, \rightsquigarrow)$ modulo $\sim$. The idea is to replace each reversible clique modulo $\sim$ by a single ball located at the center of mass of the associated cluster. The number of provably equivalent theories in the clique is indicated by the diameter of the ball. One obtains in that way a graph with 1415 vertices (blue balls of different diameters) and 4824 edges (red arrows), see Fig. 7.

We mention a number of remarkable emerging phenomena which appear when one observes the embedding of the graph $(G, \rightsquigarrow)$ modulo $\sim$.

8.1 Radial direction of implication flows

A striking observation on the picture of Fig. 7 is the existence of clearly oriented implication flows across the latent space. The mainstream direction goes along the X-axis towards equational theories with higher expectations. There is also a clear radial orientation along the Y and Z-axis towards theories with higher variance, and with different degrees of self-conjugacy.

The implication graph looks disorganised close to the source $\mathsf{Eqn}[1]$ which implies all the other theories. On the other hand, it looks very organised close to the sink $\mathsf{Eqn}[0]$ which is implied by every theory. One tentative explanation is that more equational theories $\mathsf{Eqn}[k]$ such as *idempotency* or *commutativity* can be deduced as consequences of a theory $\mathsf{Eqn}[j]$ closer to the source. These equational theories can be used in the reasoning to establish unexpected facts, and produce disorganised edges. On the other hand, when one gets closer to the sink, it becomes much more difficult to prove things, and this could explain why there are so few implication edges going against the mainstream radial orientation

8.2 Parallelism of implications

Another striking phenomenon is that many implication edges appear to be similar and parallel in the latent space. A closer inspection shows that very often, the parallel edges have similar source and target theories. An illustration is provided by the pair of parallel implications

$$\begin{array}{c}\mathsf{Eqn}[422] \Longrightarrow \mathsf{Eqn}[2851]\\ \mathsf{Eqn}[427] \Longrightarrow \mathsf{Eqn}[2861]\end{array}$$

where $\mathsf{Eqn}[422]$ and $\mathsf{Eqn}[427]$ are very similar:

$$\begin{array}{lll}\forall x.y.z. & x = x \diamond (x \diamond (y \diamond (z \diamond y)))\\ \forall x.y.z. & x = x \diamond (y \diamond (x \diamond (x \diamond z)))\end{array}$$

and for $\mathsf{Eqn}[2851]$ and $\mathsf{Eqn}[2861]$:

$$\begin{array}{lll}\forall x.y. & x = ((x \diamond (x \diamond y)) \diamond x) \diamond x\\ \forall x.y. & x = ((x \diamond (y \diamond x)) \diamond x) \diamond x\end{array}$$

We conjecture that this parallelism phenomenon is related to the Herbrand theorem in formal logic, see (Miller, 1987) and the discussion in Appendix. Note in particular that the witnesses (or expansion trees) provided by the Herbrand theory generalise the notion of *simple rewrite* elaborated and collected in Section 21 of the ET project blueprint. In that prospect, it would be worth collecting a large number of such parallel edges in the latent space and exploring how the underlying proofs are related, typically by inspecting their expansion trees, to see whether our conjecture holds. We leave that for future work.

8.3 Contrarian edges and hard implications

After observing the mainstream orientation of implication flows, we looked for atomic edges with a minority or contrarian orientation. To that purpose, we searched and collected a number of *longest paths* in the implication graph $(G, \rightsquigarrow)$ modulo $\sim$. Among these paths, we observed some unexpected behaviours like *swirls around the cliques or clusters* of provably equivalent theories. A possible explanation is that these clusters of provably equivalent theories are *active proof-theoretic areas* of the latent space, where many different equivalent

formulations of the very same concept can be established by back-and-forth: of the form:

$$\mathsf{Eqn}[j] \Longrightarrow \mathsf{Eqn}[k] \Longrightarrow \mathsf{Eqn}[j]$$

as we observed empirically.

We also considered the hard implications mentioned on Section 27 of the ET project blueprint as well as the implications with long Vampire times. We noticed empirically that a number of them had unusual and contrarian shapes as atomic edges in the latent space. At the same time, we are not entirely sure of the phenomenon at this stage, and we could not develop a sufficiently clear and complete picture of it. We thus prefer to leave this important and promising line of investigation for future work.

9 Related works

An important question underlying our work is whether (and how) the latent information provided by finite model theory is related to the proof-theoretic structure witnessed by the implication graph. From that point of view, it would be instructive to experiment with the convolutional neural network (CNN) model developed in the ET project (Bolan et al., 2024) and see whether it can be improved by integrating the latent space structure coming from Stone pairings.

Our exploration of the connection between proof theory and finite model theory is driven by the desire to understand the latent, statistical and distributional structure of deduction in machine learning, and we hope that it will bring new insights on the way large language models reason internally in their own latent space (Asperti et al., 2025; Zhou et al., 2025). From that point of view, it is very much in line and inspired by the dualistic approach to type theory and language developed in (Gastaldi and Pellissier, 2021, 2023), see also the more recent (Bradley et al., 2024).

10 Conclusion and future work

This work started as a small-scale exercise in experimental logic, building on the achievements of the Equational Theories project initiated by Terence Tao fifteen months ago, and on the exhaustive description of the implication preorder $(G, \Rightarrow)$ between 4694 equational theories established by this remarkably active and lively collaborative project.

Our plan was to proceed in two stages. In a first stage, we would define a latent space of the 4694 equational theories using ideas and methods coming from machine learning and finite model theory. Then, in a second stage, we would study the geometric (and metric) properties of the implication preorder $(G, \Rightarrow)$ of equational theories seen as *embedded* in the latent space just constructed.

The project was propelled since its origins by curiosity and exploratory joy, combined with the hope and desire of revealing meaningful hidden latent structures at the heart of mathematical reasoning. Despite the preliminary nature and technical simplicity of this work, the outcome of the experiment largely exceeds what we originally expected:

1. First, we observed with great surprise that the latent space was excessively good at detecting the signature of an equational theory, see §6.

2. Then, we measured significant differences between the length of an edge associated in the latent space to a reversible proof and to a strict atomic (non reversible) proof. From this follows that provably equivalent theories are tightly clustered together in the latent space, see §7.

3. Finally, we discovered empirically the existence of oriented *implication flows* in the latent space of equational theories, and studied the shape of hard implications, see §8.

This series of converging discoveries indicates an unexpected similitude or correspondence between the purely *model-theoretic* definition of the latent space based on Stone duality, and the purely *proof-theoretic* definition of the implication graph between equational theories. We believe that we only scratched the surface of a statistical variant (and refinement) of the celebrated Gödel completeness theorem for first-order logic, see (Heijenoort, 1967). We include in the Appendix a discussion on Herbrand theorem, which we believe should play a central role in that statistical reconstruction.

These empirical results also convey the hope of enlarging the perspective beyond algebraic reasoning, and defining a *latent space of mathematical concepts* building on the idea of a *universe of types* formulated by Martin-Löf (Martin-Löf, 1975) and recently recast as a *moduli space of types* in Voevodsky's homotopy type theory (Univalent Foundations Program, 2013).

Finally, in a dual and complementary direction, our construction of the latent space of equational theories could provide new tools and insights in the search using reinforcement learning (or other methods) for finite magmas satisfying specific algebraic laws, such as semigroups (Simpson, 2021).

11 Limitations

We introduce the latent space of equational theories and describe its main empiric properties. This is the first paper on the topic, and we thus open a new research area at the interplay of logic and machine learning. For that reason, much remains to be done. The construction of the feature space is simplistic and could be refined by taking random samples of tuples in much larger finite magmas, or on a stochastic and non-deterministic extension of the usual notion of magma. We also used a very elementary linear form of dimensionality reduction based on a PCA method. We could explore other more advanced dimensionality reduction methods such as t-SNE, see (van der Maaten and Hinton, 2008). Although this is mentioned for future work, the current paper does not include any explicit and formal measure of complexity of proofs, such as expansion trees coming from Herbrand theorem. All these limitations can be seen as opening opportunities for future works.

Acknowledgments

We are grateful to Mario Carneiro for the nice and inspiring talk he gave on the Equational Theory Project at the AITP workshop in Aussois, in September 2025. This work has received funding from the European Research Council under the European Union's Horizon 2020 research and innovation programme (Synergy Project Malinca, ERC Grant Agreement No 670624).

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A Linear regression graphs for the first and second principal components X and Y

We provide in Fig. 8 and Fig. 9 the linear regression graphs mentioned in §6.4 and in §6.5.